

Efficient Security Management for Ad Hoc Networks

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Abstract - It is necessary, though very challenging, to integrate available security technologies in a coherent manner to provide a multi-level security management in an ad hoc mobile network. This paper proposes and analyzes a policy-based network security management mechanism that consists of responsive / preemptive defensive strategies and a “Ripple Effect” security policy activating mechanism. This scheme activates different security policy modes and security levels to provide network protections efficiently.

Keywords – ad hoc, energy efficiency, polic, security, wireless;

I. INTRODUCTION

Ad hoc network nodes rely on cooperative communications to route data packets using Mobile Ad Hoc Network (MANET) routing protocols. However, these cooperative behaviors may create potential security vulnerabilities. Moreover, network nodes that have different capacities such as processing power, memory size, battery life, etc. may also have asymmetrical security-defense capabilities to against potential attacks.

On the other hand, a physical property of radio propagation should be taken into the consideration when a potential network attack happens. Generally speaking, RF signal strength fading follows the “Inverse Square law”. Therefore, each randomly located single node can only reach its neighbor nodes. Similarly, a wireless hacker can only attack or access nodes in vicinity (about one hop away). In other words, not all nodes in an ad hoc network will be attacked at the same time and at the same severity level. Consequently, not all nodes need to use the same level of security protections. Furthermore, from a network performance perspective, certain network nodes that are capable of running a strong security protection mechanism may choose not to do so, unless it is necessary, because of a possible negative impact on the system performance. The conflict of implementing strong security mechanisms and keeping a satisfactory network performance has been an issue.

There are many security techniques such as encryption, access authentication, intrusion detection / prevention, filtering, etc. that are available to defend an ad hoc network at different protection levels. Therefore, it is necessary to provide a mechanism that integrates these techniques in a coherent manner to provide a resource-efficient, dynamic, and multi-level security management. An security management should also fulfill the following goals [1]:

- The proper security mechanism needs to be carefully chosen;
- The selection of security mechanisms itself needs to be secure;
- The status (success or failure) of the selected entities needs to be indicated;
- No extra or limited state or resource are introduced.

Driven by these requirements (especially the first item) and the goal of creating a coherent integrated security protection with higher resource efficiency, we propose a *Policy-Based Security Management* mechanism to achieve security policy integration, adaptation, and activation to minimize resource consumption subject to a security protection level constraint. This mechanism is built on a policy-based network management concept. It considers the irregular nature of security attacks and ad hoc network infrastructures in conjunction with utilizing proactive and reactive ad hoc routing protocols. This proposal consists of *responsive / preemptive* defensive strategies and a “*Ripple Effect*” security policy activating mechanism. The energy efficiency improvement of the scheme is analyzed analytically and experimentally. Furthermore, this method closely abides by related IETF standards for being compatible with other network elements in practical networks.

The rest of the paper is organized as follows: Section 2 briefly introduces some wireless network security mechanisms commonly used today to show the possibility of different levels of security protections. Several terminologies of policy-based network management are defined as well. It briefly explains two MANET flat routing protocols that are used in our proposal. Section 3 provides an overview of this policy-based ad hoc network security management mechanism. Section 4 studies energy consumptions and network performance of this mechanism both analytically and experimentally. Finally, Section 5 summarizes the study and points out directions of future work.

II. BACKGROUND AND RELATED WORK

2.1 Policy-Based Network Management

The IETF Policy Framework (POLICY) Working Group has developed a policy management architecture that is considered the best approach for policy management over the Internet. The architecture could consist of the following essential components that are used in our proposal [2], [3].

- **Policy Decision Point (PDP)** - A logical entity, or a policy server, that interprets policies stored in a repository, makes policy decisions, and communicates them to a PEP.
- **Policy Enforcement Point (PEP)** - A logical entity that can apply and execute different policies. A PEP usually exists in network nodes such as routers and firewalls.
- **Local Policy Decision Point (LPDP)** – A local representative of a PDP. It exists within a network node or a device to make local policy decisions. Basic policy decisions can be programmed into this component. Any local decision or policy changes are relayed to the PDP.

The concepts and functionalities embedded in these elements are used implicitly in our proposal.

2.2 Wireless Network Security Countermeasures

Wireless network security issues have been well studied. Several sub-systems and standards [4], [5] have been proposed for encryption, authentication, intrusion detection / prevention, filtering, access control / blocking etc. in order to solve the security problems at different levels within a network paradigm. Five security feature groups have been defined by 3GPP for tackling security issues from different aspects: network access, network domain security, user domain security, application domain, and visibility / configurability [6].

Some commonly accepted or deployed countermeasures are:

- Radio link security (e.g. Temporal Key Integrity Protocol (TKIP), Advanced Encryption Standard (AES) protocol)
- Access control (e.g. 802.1x, AAA, Extensible Authentication Protocol (EAP), Smartcard, firewall)
- Packet filtering (e.g. MAC/IP address filtering, firewall)
- Traffic pattern analysis (e.g. intrusion detection)

However, most of these security research activities and practices focus on individual techniques but lack an integrated security countermeasure management.

2.3 Proactive and Reactive MANET Routing Protocols

A set of ad hoc routing protocols used in MANET is based on flat routing that consists of three types:

- Reactive - e.g. Ad hoc On-Demand Distance Vector Routing protocol (AODV)
- Proactive - e.g. Optimized Link State Routing protocol (OLSR)
- Hybrid - e.g. Temporally Ordered Routing Algorithm (TORA) that operates at both on-demand and proactive modes.

OLSR uses periodic route updates (by using the Hello message to maintain neighbor connectivity) and the Multipoint Relay (MPR) to maintain routing tables, which reduces message forwarding delays. AODV discovers routes on demand. It broadcasts local Hello messages to offer connectivity information, which is more bandwidth efficient. However, it has a relatively longer route discovery. In our proposal, the focus is on security protections over OLSR type of Ad hoc networks.

2.3 Related Work on Energy Efficient Network Security

The energy efficient secure data transmission has been studied from many interesting aspects [6]-[8]. The major efforts were focused on the energy efficiency of individual security technology. A thorough performance analysis has been given by [9] on a challenge / response based authentication at different security levels and the correspondent authentication cost. Similarly, an energy-efficient and scalable group key agreement called circular hierarchical (CH) group model was proposed by [10] and an energy efficient analysis based on communication and computational complexity was presented.

In reality, a network is usually protected by a set of security techniques controlled by a set of security policies. Therefore, it is necessary to create a mechanism that integrates security countermeasures in a coherent manner to provide a resource effi-

cient, dynamic, and multi-level protection. Section 3 presents such a mechanism proposed by our study.

III. POLICY-BASED SECURITY MANAGEMENT FOR AD HOC NETWORKS

The proposed Policy-Based Security Management for Ad hoc Wireless System mechanism consists of three distinguish strategies (reference [11] for system design details) :

- *Responsive Policy Activation strategy* – for protecting a node that is currently under an attack
- *Preemptive Policy Activation strategy* – for protecting nodes that could potentially be attacked.
- *Ripple-Ring Effect Policy Activation strategy* – for controlling attack-warning messages propagation for triggering different levels of defense techniques.

The functional diagram is shown in figure 1 with all states and transition conditions. Several assumptions are needed for confining the scope of this paper. First, it is assumed that attacks are detectable [12]. Second, each node has equipped with different types of security mechanisms and can be activated selectively [13]. Third, basic functions of policy-based network management elements are embedded in all ad hoc nodes.

3.1 Responsive Strategy

The purpose of the Responsive Strategy is to allow an ad hoc node to activate and update security policies when under an attack. These activations are based on certain pre-set or heuristically aggregated security policies. These policies are classified into different levels. For example, there are three security policy levels: (1) basic level-“b”, (2) enhanced level-“e”; and (3) the maximum level-“m”. When there is no attack, an ad hoc node is running security policies at a basic level. When attacks happen, the victim node adjusts its security level to either an enhanced level or a maximum level depending on the types of attacks. If the ad hoc node is under a virus attack, it only needs to enhance

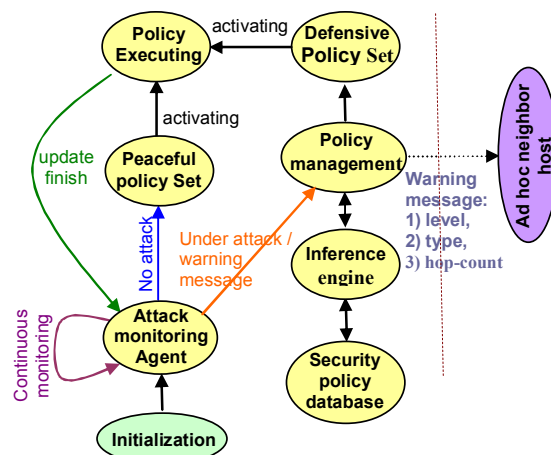


Figure 1. Responsive / Preemptive Security Management Functional Diagram

its scanning capability and updating a virus definition database. This action can be considered as an enhanced level security policy update. Compared to virus attacks, DOS attacks are much more serious. Therefore, the maximum level of the security policy set needs to be activated. Besides updating its own

security policy level, the victim node sends warning messages to its neighbors. This invokes the *preemptive strategy* of this proposal explained in Section 3.2. The concept of multi-security levels (SL_l) and multi-policy sets $P_{(j,k)}^l$ (at security level l , the k^{th} policy in policy set j) is shown in Figure 2. SL_l represents for an individual security level. In each security level (SL_l),

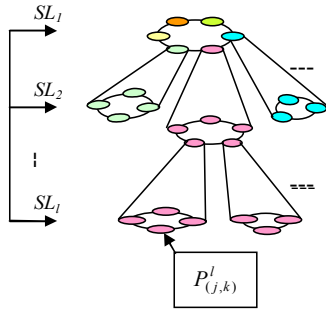


Figure 2. A Multi-level and Multi-policy Hierarchy

there are multiple security policies ($P_{(j,k)}^l$), which work together (e.g. in a ring format) to support the specific security level. Different security levels consume different level of resources.

3.2 Preemptive Strategy

When an ad-hoc node is under an attack, it updates its security policy level and sends a warning message to its neighbor nodes. When neighbor nodes receive the warning message, each of them will:

- (1) update its own security level;
- (2) relay the warning message to its own neighbors as long as the relay is within the warning distance.

The warning message is a triplet of {attack severity level, attack type, warning distance hop count}. The hop count is used in a *decay function* discussed in section 3.4. Each warning message has a message ID generated by the originator. To avoid overflowing the ad hoc network, each forwarding node keeps a cache for warning message IDs that has been forwarded. Each node will not forward the same warning message that has been forwarded by itself previously. Nodes only increase the warning message distance count when it forwards the message. To further reduce warning message flooding, it is suggested to fully utilize the ad hoc routing mechanisms, such as OLSR. For example, by using MPR, amount of duplicated warning messages will be reduced, which also avoids unnecessary collisions.

3.3 Ripple-Ring Effect Security Policy Activation

As previously mentioned, any usage of countermeasures consumes network resources and may cause certain level of performance degradation. To minimize negative impacts, the third contribution proposed is the *ripple-ring* effect security policy activating method.

When a preemptive strategy is invoked, a policy activating method is used in a ripple-ring fashion: the most powerful countermeasures will be applied to the epicenter - the “under attack” node(s); the further a node (physically counted by num-

ber of hops) is from the center, the less powerful countermeasures are activated. There is countermeasure level decreasing from the center of the attack to the nodes away from it. It is similar to throwing a stone in a pond, which causes a pattern of ripple rings that fade away from the center as the distance increases. In the proposal, the policy enforcement zones are defined according to the damage severity and the security policies needed to be applied to the zones. In this way, potential network damages could be avoided, network resources could be saved, and the counterattack response time could be reduced.

3.4 Decay Function for Ripple Ring Effect

A decay function in conjunction with the Ripple Effect strategies decides the activated security policies, so that the security level is not a constant from an attacking epicenter to the surrounding nodes on different ripple rings. The security protection could be reduced if the nodes are located within a ripple ring that is away from the epicenter. The nodes on the first several rings are the most vulnerable. These nodes need to have stronger security protections, at least as strong as the protection level of the attacked node in the epicenter. To present the security level changes among different rings, we use a decay function to describe the security level.

The decay function can be described as follows:

$\alpha = f(h)$, h : hop counts, $h \in [1, 2, 3, \dots]$ with the properties

- (1) $\lim_{h \rightarrow \infty} f(h) \rightarrow 0$;
- (2) $f(h+1) \leq f(h)$.

The decay function has a decay factor that controls the decaying speed. A sample decay function would be:

$$f(h) = \lambda \cdot d^h \quad d \in [0, 1], \quad (1)$$

where λ is a coefficient and d is a decay factor. This decay factor should be considered based on different terrain situations. For example, in a rural area, the Friis free space propagation

model can be used: $P_R = P_T G_T G_R \left(\frac{\lambda}{4\pi d}\right)^2$ [14]. In an urban area, Okumura's model is applicable:

$$L_{50}(dB) = L_F + A_{mu}(f, d) - G(h_{te}) - G(h_{re}) - G_{AREA}.$$

Please refer to [14] for the details regarding parameters. The rationality of using different decaying speeds results from the random congregation of ad hoc nodes. For example, in an urban situation, neighboring nodes are physical located closer, which could be closer to the attacked node. Hence, we should reduce the decaying speed. In contrast, in the rural environment, the security level decaying could be faster since neighbor nodes are located relatively far apart.

IV. ENERGY EFFICIENCY STUDY

4.1 Security Policy Driven Resource Consumption

In this analysis, we derive a general formula for resource consumption of an ad hoc wireless network that uses the proposed mechanism. We consider that, at any moment t , the resource consumption $R_i(t, \phi, \eta, M)$ of a node i is influenced by the following elements:

M_i – the number of activated policy at time t on node i . The policies that are used for controlling authentication, intrusion detection, virus scanning, encryption, decryption, anti-jamming, etc. consume different levels of resources such as computing cycles, memory, and battery. M_i is defined based on the decay function, security level, and activated policy set. That is: for the l^{th} level of security protection, the activated policy sets $P_{(i,k)}^l$ (please refer to Figure 2, K is in the same set (color), j is in different set (color)), and the location of the node h_i :

$$M_i = \sum_{j=1}^J \sum_{k=1}^K f(\alpha(h_i), P_{(j,k)}^l) \cdot (1)$$

$\eta_i(M_i, t)$ – the accumulative resource consumption due to communication. It represents the resource used by message exchanges generated by types and levels of the security protection, e.g. the energy that is used for the challenge / response [9], key exchange [10], policy lookup, etc. These messages have different sizes and different dialog durations (depending on each protocol), which consumes different levels of resources. Therefore, it is a function of M_i and t .

$$\eta_i(t) = \sum_{n=1}^{M_i} \eta_n(t), \quad (2)$$

where $\eta_n(t)$ is the resource used by messages of node i for n^{th} protocol (or security means) at the moment t .

$\Phi_i(M_i, t)$ – the accumulative resource consumption due to computing. It reflects the resource consumed by computation involved in, e.g. encryption, decryption, etc. Similarly, we have:

$$\phi_i(t) = \sum_{q=1}^{M_i} \phi_q(t)^\dagger, \quad (3)$$

where $\Phi_q(t)$ is the resource used by computing for q^{th} algorithm (or security means) of node i at the moment t .

All of the above elements are functions of t . As time progresses, each security policy may at a different phase and use different levels of resources. Finally, by integrating the above items over the time duration (from t_0 to t_1) of total running policy, a total resource consumption of a node i is:

$$R_i = \int_{t_0}^{t_1} r_i \left[\sum_{n=1}^{M_i} \eta_n(t), \sum_{q=1}^{M_i} \phi_q(t) \right] dt \cdot (4)$$

Then, for a network that has a decay distance D is under an attack, its total resource consumption (using the proposal) is:

$$\bar{R}_{Ripple} = P_a \left\{ R_0 + \sum_{r=1}^D N_r R_r \right\} + \sum_{r=D+1}^R N_r R_B + (1 - P_a) \sum_{r=0}^R N_r R_B, \quad (5)$$

where R is the ring number in a “ripple”, P_a is the attack probability, N_r is the number of ad hoc nodes in the r^{th} ring. The decay function of defensive strategy helps to guarantee the following conditions:

$$R_0 > R_1 > R_2 > \dots > R_D > R_B, \quad (6)$$

[†] The formulas of $\eta_i(t)$ and $\Phi_k(t)$ are system dependent.

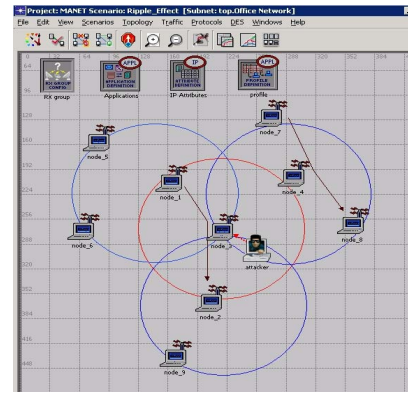


Figure 3. An Ad Hoc Network Simulation Model

where R_0 is the attack center (ground zero) node resource consumption, R_1 is for ring 1, and so on, R_D is for the end of a decay distance, and R_B is for the basic level nodes.

4.2 An Example and Simulation Results

Using a three-tiers (rings) topology shown in Figure 3 as an example ($D = 3$, $R = 4$), we illustrate the network resource efficiency improvement as the followings:

$$\bar{R}_{Ripple} = P_a \left\{ R_0 + \sum_{r=1}^3 N_r R_r \right\} + \sum_{r=4}^4 N_r R_B + (1 - P_a) \sum_{r=0}^4 N_r R_B \cdot (7)$$

Each R_r is the resource consumption presented in (4). R_0 is the resource consumption for the counter attack of the maximum level attack, R_1 is for the enhanced level-1, and so on. R_B is the resource consumption for an ad hoc node without suffering an attack. N_1 represents nodes under highest protection, the same as the attacked node. N_2 represents the nodes with the second tier (level-2) protections based on the decay function, and so on. In this example, N_4 are the nodes without suffering attacks.

Comparatively, the traditional defense resource consumption can be presented as:

$$\bar{R}_{traditiond} = P_a \left\{ R_0 + \sum_{r=1}^3 N_r R_0 \right\} + \sum_{r=4}^4 N_r R_0 + (1 - P_a) \sum_{r=0}^4 N_r R_0 \cdot (8)$$

The first part of equations (7) and (8) are the same, both equal to P_a times R_0 . This represents the resource used for the attacked nodes. The remaining parts of (7) and (8) are different. For the traditional protection mechanism, all nodes in an ad hoc network are updated to the level of protections that the victim node has. In comparison, the Ripple Effect updates a node security level base on the distance and an output from the decay function. Therefore, the resource requirements for (7) are much less than the one in (8). It shows that the ripple-effect defense strategy uses less resources, which could yield a higher performance for data process and communications.

The computer simulation has been done by using OPNET. The network configuration is shown in Figure 3. The experiment is based on an ad hoc network in a 500x500 m² area with nine mobile nodes and one attacking node. These nodes are equipped with 802.11b wireless cards and work in the ad hoc mode using the OLSR routing protocol. Each node is capable of reaching 100 meters. The total simulation time is 500 seconds. When the center node, node_3, is under an attack, it broadcasts

a warning message to the second tier nodes, which are node_1, node_2, and node_4 marked in a “red” circle. The second tier nodes then forward the warning message to the third tier nodes, they are node_5, node_6, node_7, node_8, and node_9 marked in “blue” circles. All nodes will update their security levels accord to the Ripple Effect Algorithm.

In order to demonstrate the resource utilization at different security levels, we assume that the maximum, enhanced, and basic levels use different percentage of network resources from the highest to the lowest.

We apply video traffic over this ad hoc network. There are two identical flows of low rate video traffic in the network. One is from node_1 to node_2 routed via node_3, and the other is from node_7 to node_8 routed via node_4. Both traffic start from the 50th second with a data rate of 10 packets per second. An maximum attack, e.g. a DOS, is initiated by the attacker node to against the middle node, node_3. The attack starts at the 100th second and finishes at the 200th second. Notice that node_3 is the direct target for the DOS attack. This attack triggers a maximum security level protection on node_3. It is the first tier node that uses more resources than nodes in other tiers. Node_3 sends the warning messages to its neighbors. As a comparison, the node_4 is a second tier node that is at the enhanced level protection. Its dedicated resources to security is much less than tier one node Node_3. Figure 4 shows the simulation results. The first curve is the video traffic source. The second curve is the DOS attack traffic. The third curve is the first video stream received at the receiver (node_2). Significant packet loss during the time of DOS attack (100th – 200th second) is illustrated. As a comparison, the fourth curve shows the video stream received at the node_8, which has little impact from the attack. The reason is that the node_3 uses more network resources to protect itself than the node_4 does. The differences of available node resources have significant impact on the network

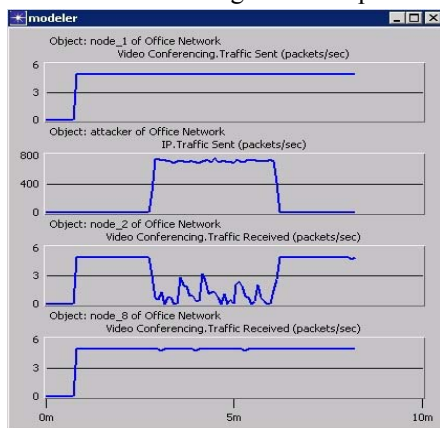


Figure 4 Throughput Comparison of Ripple Effect Algorithm

service performance. These simulation results have shown that the Ripple Effect based strategy has better performance than conventional security mechanisms, which is consistent with the analytical result.

V. CONCLUSION

In this paper, we have discussed the security challenges imposed on ad hoc networks. A *Policy-Based Security Manage-*

ment for Ad Hoc Wireless System mechanism has been proposed. The performance analysis and experiment have been presented. We believe that the integrated and adaptive nature of this mechanism could bring more robust defenses to ad hoc networks. This mechanism can also be applied to sensor networks.

However, there are more complicated cases that should be further investigated. For example, the “ripple” superposition issue when different nodes under different attacks is worth to study. A more finer granularity in the policy sub-set within one policy type can be further developed to improve the defense / energy sensitivity. Further performance analyses, in terms of counterattack response time, node survivability, and performance fluctuation will also be continued in the future.

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